

**BRAC UNIVERSITY**  
**Department of Computer Science and Engineering**

Examination : Semester Midterm  
 Duration: **1 Hour 20 Minutes**

Semester: Fall 2025  
 Full Marks: **50**

CSE421 / EEE465 : Computer Networks

Answer **ALL** questions. (**Pages: 2**)

Figures in the right margin indicate marks.

|              |            |                 |
|--------------|------------|-----------------|
| <b>Name:</b> | <b>ID:</b> | <b>Section:</b> |
|--------------|------------|-----------------|

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                  |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| <b>Q1</b><br><b>[CO1]</b> | A software bug causes encrypted data to be delivered directly to the application layer without establishing a session. Identify the OSI layers that have been bypassed or mis-sequenced, and describe the problems Bob would observe at the application side.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>4</b>                         |
| <b>Q2</b><br><b>[CO2]</b> | <p>A news website stores all article metadata under <code>/articles/{id}/info.json</code> on its origin server. A university network uses a proxy cache to speed up web access for students.</p> <p>At 2:00 PM, the content team updates <code>/articles/55/info.json</code> directly on the origin server. The proxy still contains an older cached version from before the update.</p> <p>At 2:06 PM, a student submits a comment on the same article using a form that sends data to <code>/articles/55/comments</code>.</p> <p><b>I.</b> State which HTTP method the browser will use to submit this comment. Briefly explain.</p> <p>At 2:07 PM, another student requests: <code>GET /articles/55/info.json</code> through the proxy.</p> <p><b>II.</b> State which version of the file the proxy will return to the student, and why?</p> | <b>2</b><br><b>+</b><br><b>4</b> |
| <b>Q3</b><br><b>[CO2]</b> | <p>FastFoodHub is an online food ordering website. When customers add items to their cart and refresh the page, the cart items stay saved. But some customers complain that on another day, when they use a different device, their cart becomes empty even though they are logged into the same account.</p> <p><b>I.</b> What web technology is being used to keep the cart items saved on the same device?</p> <p><b>II.</b> Why do the cart items not appear when the customer switches to a different device?</p>                                                                                                                                                                                                                                                                                                                          | <b>2</b><br><b>+</b><br><b>2</b> |
| <b>Q4</b><br><b>[CO2]</b> | <p>Jannat tries to open <code>research.university.edu</code>, but her browser keeps showing an error that the site cannot be reached. Her teammate, who is using a different network, can open the same website instantly. Later, Jannat clears her DNS cache, and after that, the website loads normally on her device.</p> <p><b>I.</b> Briefly explain how DNS resolution enables access to a website, and the reason why clearing the DNS cache solved Jannat's problem.</p> <p><b>II.</b> Discuss why recursive resolution is convenient for clients while iterative resolution benefits DNS infrastructure.</p>                                                                                                                                                                                                                           | <b>3</b><br><b>+</b><br><b>3</b> |
| <b>Q5</b><br><b>[CO2]</b> | Anika is attending an online lecture with 120 participants while also downloading a 200 MB course file in the background. During the lecture, a few words drop out because of weak Wi-Fi, but the session continues without stopping. The file download takes longer but finishes perfectly                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>4</b><br><b>+</b><br><b>2</b> |

|                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                       |                                                                  |   |    |    |    |      |    |    |    |      |    |    |    |    |            |   |   |   |   |    |    |    |      |    |    |   |      |   |   |                                                                  |
|------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|---|----|----|----|------|----|----|----|------|----|----|----|----|------------|---|---|---|---|----|----|----|------|----|----|---|------|---|---|------------------------------------------------------------------|
|                                                                  | with no missing data.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                       |                                                                  |   |    |    |    |      |    |    |    |      |    |    |    |    |            |   |   |   |   |    |    |    |      |    |    |   |      |   |   |                                                                  |
|                                                                  | <div>I. If the online lecture used the same protocol as the file download, describe what would happen to the live session and explain why this would be a problem.</div> <div>II. Explain why the file took longer time to download.</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                       |                                                                  |   |    |    |    |      |    |    |    |      |    |    |    |    |            |   |   |   |   |    |    |    |      |    |    |   |      |   |   |                                                                  |
| <div>Q6</div> <div>[CO3]</div>                                   | <div>Consider a TCP connection using slow start and congestion avoidance.</div> <div><div>The initial slow-start threshold (sssthresh) is 15 MSS.</div><div>The congestion window (cwnd) changes over the rounds as shown:</div></div> <table><tr><td>Round</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td><td>11</td><td>12</td><td>13</td></tr><tr><td>cwnd (MSS)</td><td>1</td><td>2</td><td>4</td><td>8</td><td>15</td><td>16</td><td>17</td><td>18→9</td><td>10</td><td>11</td><td>?</td><td>13→1</td><td>2</td><td>4</td></tr></table> <div><div>I. Identify why cwnd = 15 at round 4.</div><div>II. What will be the cwnd at round 10?</div><div>III. Between Round 7 and 11, which one indicates more severe congestion?</div></div> | Round                                                                                                                                                                                                                                                                                                 | 0                                                                | 1 | 2  | 3  | 4  | 5    | 6  | 7  | 8  | 9    | 10 | 11 | 12 | 13 | cwnd (MSS) | 1 | 2 | 4 | 8 | 15 | 16 | 17 | 18→9 | 10 | 11 | ? | 13→1 | 2 | 4 | <div>2</div> <div>+</div> <div>2</div> <div>+</div> <div>2</div> |
| Round                                                            | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 1                                                                                                                                                                                                                                                                                                     | 2                                                                | 3 | 4  | 5  | 6  | 7    | 8  | 9  | 10 | 11   | 12 | 13 |    |    |            |   |   |   |   |    |    |    |      |    |    |   |      |   |   |                                                                  |
| cwnd (MSS)                                                       | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2                                                                                                                                                                                                                                                                                                     | 4                                                                | 8 | 15 | 16 | 17 | 18→9 | 10 | 11 | ?  | 13→1 | 2  | 4  |    |    |            |   |   |   |   |    |    |    |      |    |    |   |      |   |   |                                                                  |
| <div>Q7</div> <div>[CO3]</div> <div>[CO3]</div> <div>[CO2]</div> | <div>Asif loads an HTML page in Google Chrome over a persistent HTTP connection (TCP termination time and the third phase of handshaking are negligible). The page contains 11 objects in total, including the base HTML file. The size of each object is 4 MB. The total round-trip time to load the page is 455 ms. Note that the TCP RTT is twice the per-object HTTP request-and-response RTT. The server’s bandwidth is 16 Mbps.</div> <div><div>I. Calculate the TCP RTT and the per-object HTTP request response RTT.</div><div>II. Calculate the total page load time.</div><div>III. If the page had 20 objects instead of 11, explain which part of the delay would increase the most in persistent HTTP, and why.</div></div>                                                                         | <div>3</div> <div>+</div> <div>3</div> <div>+</div> <div>3</div>                                                                                                                                                                                                                                      |                                                                  |   |    |    |    |      |    |    |    |      |    |    |    |    |            |   |   |   |   |    |    |    |      |    |    |   |      |   |   |                                                                  |
| <div>Q8</div> <div>[CO3]</div> <div>[CO3]</div> <div>[CO2]</div> | <div>At a certain point, the client transmits segment C1 with a sequence number of 1996 and an acknowledgment number of 2025. The client’s rwnd is 12000 bytes, and the server’s rwnd is 10000 bytes before segment C1 was sent. The data size of the C1, C2, S1, S2, S3, S4 are 1500, 3500, 2500, 1000, 3000, and 2000 bytes, respectively.</div> <div><div>I. Determine whether Selective Repeat or Go-Back-N is used. Note that when the client sends the second C2 segment, the advertised rwnd is 8500 bytes.</div><div>II. Find the sequence and acknowledgment numbers of the S4 segment.</div><div>III. Explain why the client retransmitted the C2 segment, even though the RTO for C2 did not expire.</div></div>                                                                                      | <div><div><div>Client</div><div>Server</div><div><div><div>C1</div><div>S1</div><div>S2</div><div>C2</div><div>S1</div><div>S2</div><div>S3</div><div>S4</div><div>C2</div></div><div><div><div>RTT Start</div><div>RTT End</div><div>RTT Start</div><div>RTT End</div></div></div></div></div></div> | <div>3</div> <div>+</div> <div>4</div> <div>+</div> <div>2</div> |   |    |    |    |      |    |    |    |      |    |    |    |    |            |   |   |   |   |    |    |    |      |    |    |   |      |   |   |                                                                  |

